



IOTRICITY S3

Team Name: Assymetricals

**Problem Statement:
EDGE-ML Based Smart Fire Extinguisher.**

Problem identification & analysis

PROBLEM:

Fire spreads faster than conventional systems can respond.

- Rapid fire propagation
- Conventional sprinklers flood entire zones
- Collateral damage to equipment & infrastructure
- No precise information about *where* the flame is

THE GAP:

Conventional Systems,

Fire → Detection → Entire Zone Suppression

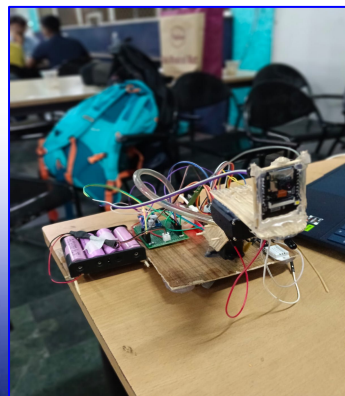
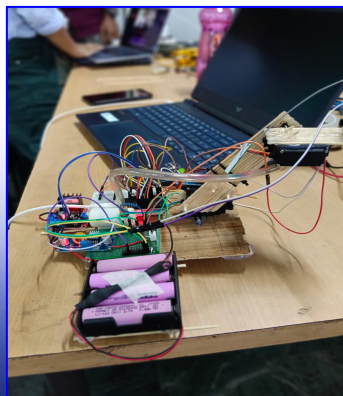
Our solution,

Fire → Vision → Verification → Precise Suppression

The missing capability: *spatial awareness*.

OUR SOLUTION:

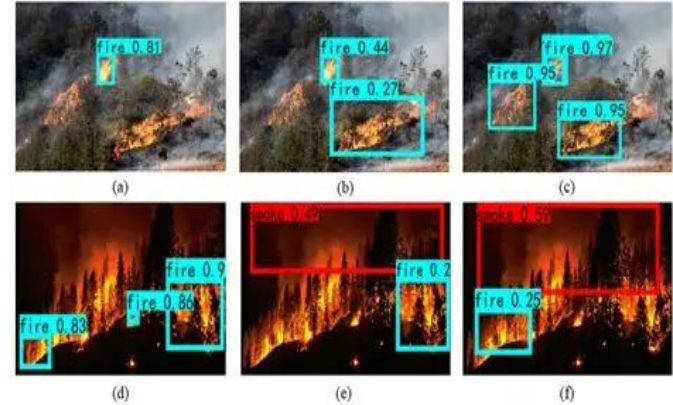
Edge-ML based fire-monitoring and targeting system that has vision-guided autonomous fire extinguisher that will detect flames with YOLOv8 combined with smoke detection with ultrasonic ranging. When a potential fire is detected, two servo-mounted water cannons sweep the area, identify the closest target angle, and hold that position while the alert indicators remain active.



Proposed Technical Approach

Working Principle:

- Sensors keep checking distance and smoke.
- If fire is detected or smoke is high, the alarm turns on.
- The servos sweep left to right and find the direction with the closest object.
- They lock onto that direction, and the OLED shows the status.
- It re-scans every 3 seconds until the threat is gone.



Working Flow:



Technology Stack

Hardware Components (With Price):

- **Arduino Uno:** main controller, about ₹700
- **2× Servo motors :** aim the water cannon, about ₹300
- **HC-SR04** ultrasonic sensor: measures distance, ₹80
- **MQ-2 gas/smoke sensor:** confirms smoke, about ₹150
- **SSD1306 128×64 OLED:** shows status, about ₹300
- **Buzzer (5 V):** audible alarm, about ₹20
- **Red and green LEDs with 2× 1 kΩ resistors:** threat and normal indicators, about ₹20
- **10 kΩ potentiometer:** adjusts smoke threshold, ₹30
- **Breadboard and jumper wires:** connections, about ₹150

Software Used:

- **Embedded C (*Arduino IDE*):** firmware and state machine
- **Wokwi:** circuit simulation
- **YOLOv8 (quantized):** flame detection on the edge
- **Adafruit SSD1306/GFX and Servo libraries:** OLED and servo control
- **Serial Monitor (115200 baud):** telemetry and debugging

Check The Repository:

<https://github.com/rajdeep13-coder/Smart-Fire-Extinguisher>

Wokwi Simulation:

<https://wokwi.com/projects/476314925594840065>



Feasibility & Viability

TECHNICALLY FEASIBLE:

- Sensing & scanning logic simulated in **Wokwi**
- Lock-on & display logic implemented
- **YOLOv8** detection pipeline ready
- Designed for **edge inference** with no cloud dependency

LOW-COST:

₹1,750–₹2,000

Prototype BOM

Arduino • Servos • Sensors • OLED • Camera •
Edge compute • Pump/Valve • Driver

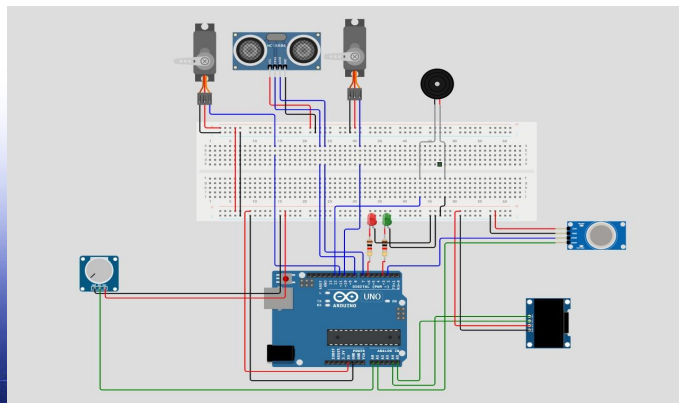
The cost is approximate.

RAPID DEPLOYMENT:

3–4 hrs*

**Estimated initial hardware
integration**

Simulation → Wiring → Calibration
→ Testing

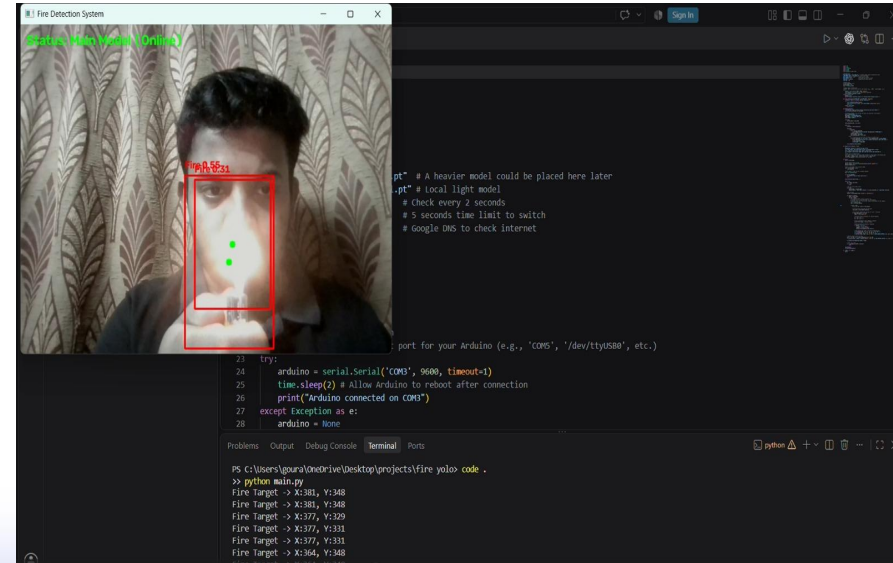
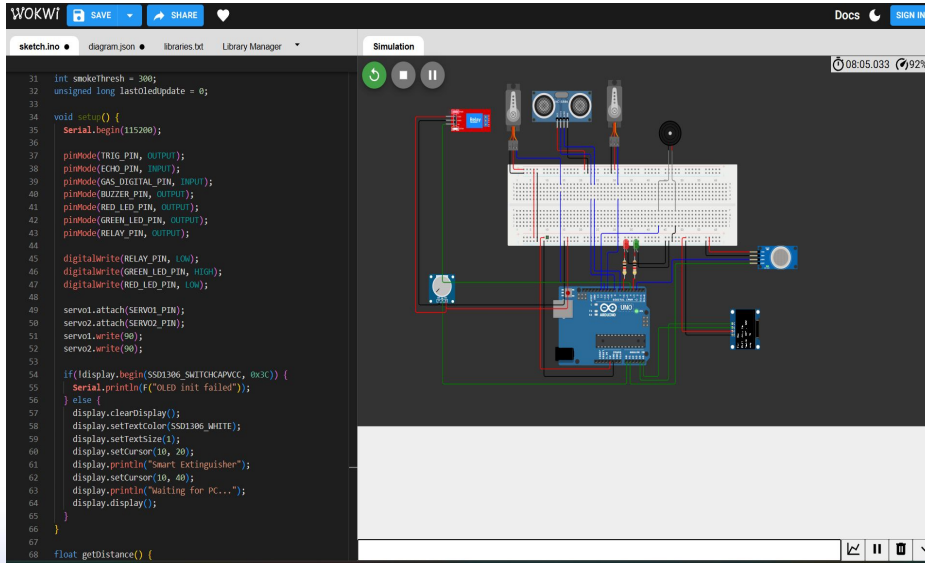


Challenges	Mitigation
False positives	Sensor Fusion
Aiming error	Camera calibration
Pump current spikes	flyback protection

Impact & Benefits

- **Target Beneficiaries:** Industrial facilities, warehouses, data centers, battery storage sites, commercial kitchens and labs.
- **Problem Solving:** Replaces delayed, area-wide sprinklers with instant, targeted suppression.
- **Key Benefits: Speed:** sub-second response.
 - **Safety:** acts before fire spreads.
 - **Cost:** conserves water and limits equipment damage.
 - **Reliability:** works offline and during power/network disruption.
 - **Accuracy:** dual confirmation cuts false alarms.
- **Expected Impact:** Less property loss, fewer injuries, and faster containment before main systems deploy.
- **Impact Measurement:** Response time from detection to discharge, aiming accuracy, false-alarm rate, and water used versus a sprinkler.
- **Limitations & Future Scope:** The simulation uses a single-axis sweep, and the camera, YOLOv8 and pump are not yet integrated.
 - It needs formal safety validation before real deployment.
 - Next steps: full dual-axis targeting, domain-specific suppression media, and human override.

The Simulations:



Youtube Link: <https://www.youtube.com/watch?v=LdXi64k5plc>